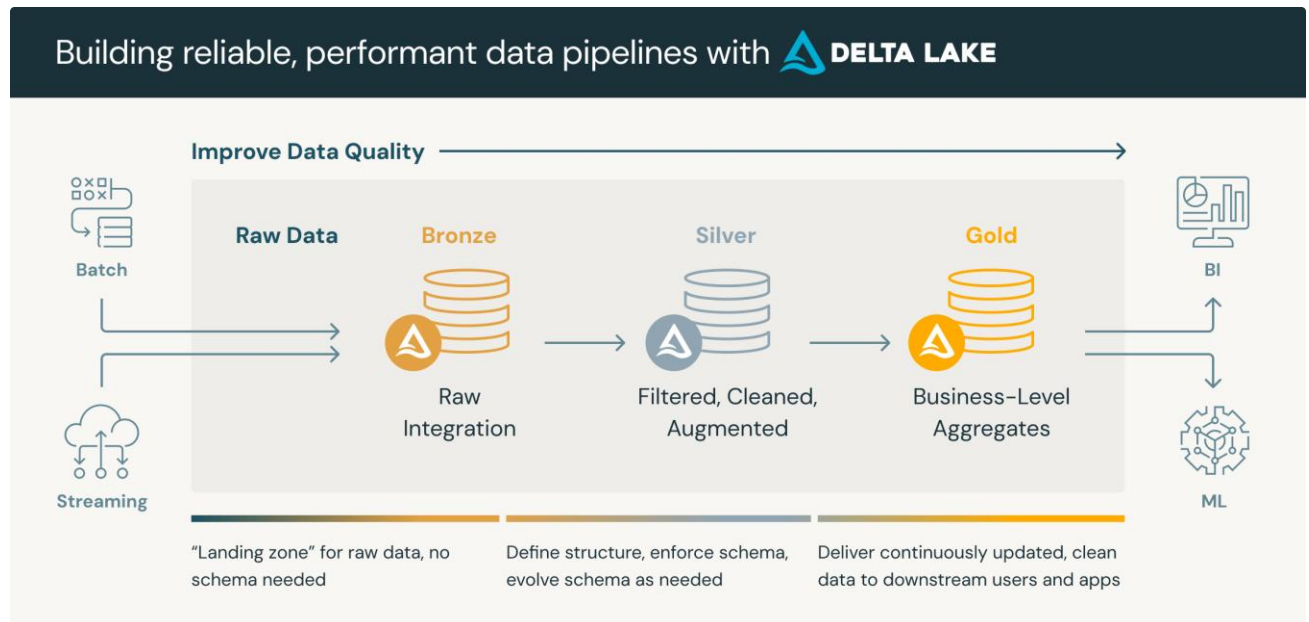


Databricks 14 Days AI Challenge

Day 6: Medallion Architecture



1. What is Medallion Architecture

It's a layered data design pattern: **Bronze** → **Silver** → **Gold**

Each layer has a purpose. You never skip layers. You never mix responsibilities.

Think of it like:

- Bronze = raw reality
- Silver = clean truth
- Gold = business value

Bronze Layer (Raw Data) = audit Layer

What it contains

- Raw ingested data
- No transformations
- No business logic
- Only minimal formatting fixes

Exactly what the source system produced. Example:

- Raw clickstream

- Raw CSV from vendors
 - Raw API payload
 - CDC feed
-

What you DO in Bronze

- Ingest data
- Add ingestion timestamp
- Add source file metadata
- Store as Delta
- Append-only

```
python

raw_data.write.format("delta") \
    .mode("append") \
    .saveAsTable("ecommerce.bronze_events")
```

Silver Layer (Cleaned + Enriched)

This is where real engineering happens.

What Silver contains

- Cleaned data
- Deduplicated records
- Proper data types
- Joined dimension data
- Business-ready fields
- Slowly Changing Dimensions (Merge)

This is where:

- invalid rows removed
- null handling
- schema corrections
- surrogate keys added

```
python
```

```
silver = bronze \
    .filter("user_id IS NOT NULL") \
    .dropDuplicates(["event_id"]) \
    .withColumn("event_time", F.to_timestamp("event_time"))
```

Gold Layer (Aggregated / Business Layer)

This is what:

- BI dashboards
- Analysts
- ML features
- Reporting tools

What Gold contains

- Aggregated tables
- KPI metrics
- Revenue summaries
- Customer-level features
- Category-level performance

Example:

```
gold = silver \
    .groupBy("category_code") \
    .agg(F.sum("price").alias("total_revenue"))
```

Stored as:

```
python
```

```
gold.write.format("delta") \
    .mode("overwrite") \
    .saveAsTable("ecommerce.gold_category_revenue")
```

How the layer flows:

BRONZE → SILVER → GOLD

Each layer reads only from the previous layer.
Never directly from raw source to Gold.

2. Best Practices for each layer

Bronze Best Practices

- Append-only
 - Partition by ingestion date
 - Store metadata (file name, load time)
 - No deletes
 - Keep schema flexible (schema evolution allowed)
-

Silver Best Practices

- Enforce schema
- Deduplicate records
- Handle late-arriving data
- Use MERGE for updates
- Add data quality checks

Gold Best Practices

- Keep tables small and fast
- Pre-aggregate heavy computations
- Partition wisely
- Optimize + ZORDER
- Designed for consumption, not engineering

3. Incremental Processing Patterns

This is where most interviews go deeper.

You don't reprocess everything daily. You process only new data.

Pattern 1: Append-only Incremental

Daily ingestion: Silver only reads new partitions

```
Bronze.append(new_data)
```

Pattern 2: MERGE-based Incremental (CDC)

If source sends updates:

```
sql

MERGE INTO silver_table
USING new_data
ON silver_table.id = new_data.id
WHEN MATCHED THEN UPDATE
WHEN NOT MATCHED THEN INSERT
```

This avoids full refresh.

Pattern 3: Watermark-based (Streaming)

Track last processed timestamp:

```
.where("event_time > last_processed_time")
```